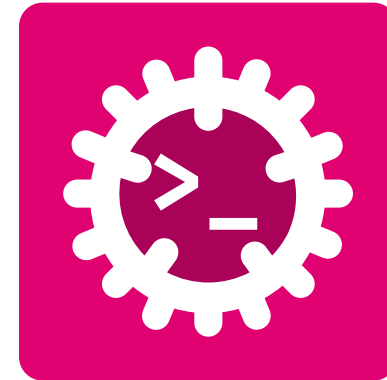


Advanced Programming

Degub // Testing // Documentation



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Debug / Testing / Documentation

Ship code that works - and prove it



The safety net of a healthy Rust project - each layer catches a different class of mistake.

- **Unit tests** - `#[test]` functions, run by `cargo test`
- **Integration & doc tests** - the `tests/` folder and runnable examples in docs
- `cargo check / clippy` - fast type-check and lints
- `cargo fmt` - one canonical style
- **Error handling** - `Result / Option`, no surprise crashes
- **Logging & debugging** - see what the program actually did
- **Profiling** - measure before you optimise
- **CI/CD** - run **all** of the above automatically on every push

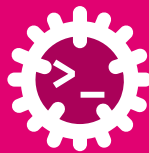


The goal: catch a mistake **as early and as cheaply** as possible - ideally at compile time.



Testing

Prove the code does what you claim



A **unit test** checks **one** piece of code in **isolation** - including the **edge cases**.

```
// the code under test
pub fn parse_temperature(s: &str) → Option<i32> {
    s.trim().trim_end_matches("°C").parse().ok()
}

#[cfg(test)]                // compiled only for `cargo test`
mod tests {                 // a *submodule* for the tests
    use super::*;           // bring the parent module into scope

    #[test]                 // a *unit test* function
    fn plain_number() {
        assert_eq!(parse_temperature("20"), Some(20));
    }

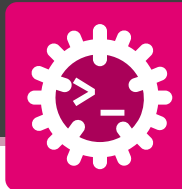
    #[test]
    fn strips_unit_and_spaces() {
        assert_eq!(parse_temperature(" 20°C "), Some(20));
    }

    #[test]
    fn rejects_garbage() {
        assert_eq!(parse_temperature("hot"), None);
    }
}
```

- lives **next to the code**, in `#[cfg(test)]` mod tests
- `#[test]` marks each test function
- the rhythm: **arrange** ⇒ **act** ⇒ **assert**
- test the **happy path**, the **edge**, and the **failure**
- run them all with `cargo test`



A failure prints the **file, line** and **both values** - it points right at the bug.



The **same** function, tested two more ways - both run under cargo test.

Integration Tests

files in tests/, exercise the crate **from outside**

- only public API is visible

```
// tests/temperature.rs
use weather::parse_temperature;

#[test]
fn reads_celsius() {
    assert_eq!(parse_temperature("20°C"), Some(20));
    assert_eq!(parse_temperature("nope"), None);
}
```

Doc tests

Code in `///` examples is **compiled and run** ⇒
documentation that **can't go stale**

```
/// Parses a temperature like `20°C`. DOC test
///
/// # Examples
/// ```
/// use weather::parse_temperature;
/// assert_eq!(parse_temperature("20°C"), Some(20));
/// ```
pub fn parse_temperature(s: &str) → Option<i32> {
    // ...
}
```




An assertion says what **must** be true - and sometimes the **correct** behaviour is to **panic**.

```
// °F → °C, rejecting impossible input
fn f_to_c(f: f64) → f64 {
    assert!(f ≥ -459.67, "below absolute zero");
    (f - 32.0) * 5.0 / 9.0
}

#[test]
fn freezing_and_boiling() {
    assert_eq!(f_to_c(32.0), 0.0); // = (both shown on fail)
    assert_ne!(f_to_c(212.0), 0.0); // ≠ (it is 100.0)
}

#[test]
#[should_panic(expected = "below absolute zero")]
fn rejects_impossible() { f_to_c(-500.0); } // pass if it panics

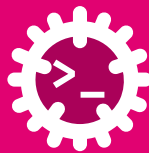
#[test]
fn parses_ok() → Result<(), String> {
    let t = parse_temperature("20°C").ok_or("bad")?;
    assert_eq!(t, 20); // a test may return Result
    Ok(())
}
```

- `assert_eq!` / `assert_ne!` print **both values** on failure
- `assert!(cond, "msg")` - assert a bool, with a **message**
- `#[should_panic]` - the test **passes only if it panics**
- a test may **return Result** - ? inside, an `Err` fails it
- filter: cargo test freezing runs all function with the word freezing in the function name.
⇒ Here only `freezing_and_boiling()`



Code Quality & CI

Automate the boring checks



Three commands that catch **most** mistakes before a single test runs.

```
cargo fmt      # one canonical style
cargo check    # type-check only, no binary
cargo clippy   # ~750 lints, catches foot-guns
# auto-fix where possible
cargo fmt --all
cargo clippy --fix
```

Rust **fmt** can be configured with `rustfmt.toml` - but the **default** is good enough for most projects. Just run it **on save** in your editor and forget about it.

```
edition = "2024"
tab_spaces = 2
max_width = 160
reorder_imports = true
```

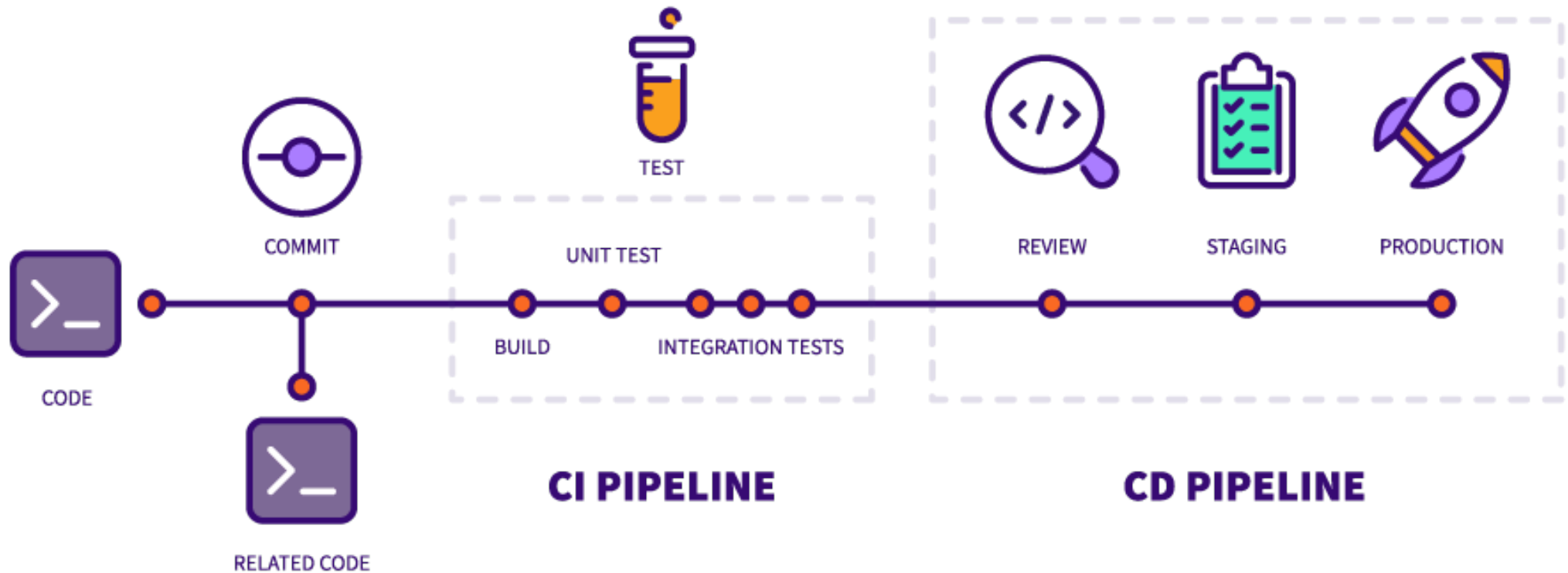
- **fmt** - rustfmt; run it **on save** and end style arguments
- **check** - compiler **front-end** only → fast feedback loop
- **clippy** - idiomatic-Rust **linter**; explains **why** and suggests a fix



In **CI**, make warnings fatal: `cargo clippy -- -D warnings`.



Run every check **automatically** on each **push / pull-request**





```
# .github/workflows/ci.yml
on: [push, pull_request]
jobs:
  fmt:
    - run: cargo fmt --all -- --check
  clippy:
    - run: cargo clippy -- -D warnings
  test:
    - run: cargo test --verbose
  build:
    - run: cargo build --verbose
```

- the **same** commands you run locally - now enforced for **everyone**
- four parallel jobs: **fmt** · **clippy** · **test** · **build**
- on **every push & PR** to main



Green check = formatted, lint-clean, builds, **all tests pass.**

The screenshot shows the GitHub Actions interface for a workflow named `ci.yml` triggered by a push to the `main` branch. The status is **Success**. The summary shows a total duration of **1m 38s**. A list of jobs on the left includes Test Suite, Rustfmt, Clippy, Build Check, and cross-platform builds for Ubuntu, Windows, and macOS, all with green checkmarks. The right panel shows a detailed view of the `ci.yml` workflow, listing the duration of each job: Test Suite (50s), Rustfmt (8s), Clippy (24s), and Build Check (50s). Below this, a 'Matrix: Cross-platform build' section indicates that 3 jobs completed successfully.

Job	Duration
Test Suite	50s
Rustfmt	8s
Clippy	24s
Build Check	50s

`unwrap()` was a mistake



Handling Failures

When things go wrong, gracefully



Result<T, E> and Option<T> make failure **part of the type** (see **Imperative Programming**).

Method	What it does	Use when
Propagate / handle		
?	hands the Err/None to the caller	the caller should decide
match / if let	branch on each case explicitly	you handle both paths here
Fallback - never panics		
unwrap_or(x)	the value, or x	you have a cheap default
unwrap_or_else(f)	the value, or f()	the default is expensive to build
unwrap_or_default()	the value, or T::default()	0 / "" / empty is fine
Panic - give up		
unwrap()	the value, or panic	tests / “can’t happen”
expect("msg")	the value, or panic w/ msg	document the invariant
panic!()	abort the thread	unrecoverable bug



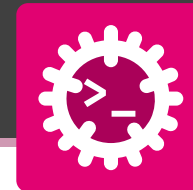
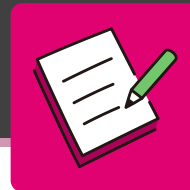
```
use std::fs;
fn load_port(path: &str) → Option<u16> {
    // ? propagates None upward if read fails
    let text: String = fs::read_to_string(path).ok()?;
    // ? propagates None upward if parsing fails
    let port: u16 = text.trim().parse().ok()?;
    Some(port)
}

fn main() {
    // 1. Fallback - never panics: use 8080 if loading fails
    let port: u16 = load_port("app.toml").unwrap_or(8080);
    // 2. Handle both arms explicitly: success and failure
    if let Some(p) = load_port("app.toml") {
        println!("listening on {p}");
    } else {
        eprintln!("config problem, using defaults");
    }
    // 3. Fallback with a computed/expensive default
    let port: u16 = load_port("app.toml").unwrap_or_else(|| {
        eprintln!("falling back to default port");
        8080
    });
}
```

```
// 4. Fallback to the type's default value
let port: u16 = load_port("app.toml").unwrap_or_default();
// 5. Last resort in a prototype: panic with a message if
// config is missing/invalid
let port: u16 = load_port("app.toml").expect("config
required");
// 6. Unrecoverable state: abort explicitly
if port == 0 {
    panic!("port 0 is never valid");
}
}
```

In the example

- ? - bubble the error **up**
- unwrap_or - safe default 8080
- if let / match - handle **both** arms
- expect - **panic** w/ message
- panic! - **abort** the thread



The code is given. On each `// TODO` line, add the **right method** from the failure toolbox - each line uses a **different** one (`?`, `unwrap_or`, `unwrap_or_else`, `unwrap_or_default`, `expect`).

```
use std::num::ParseIntError;

// parse_port returns Result<u16, ParseIntError>
fn parse_port(s: &str) → Result<u16, ParseIntError> {
    s.trim().parse()
}

// TODO 1: propagate the Err to the caller (one operator)
fn sum_ports(a: &str, b: &str)
    → Result<u16, ParseIntError> {
    let x = parse_port(a)/* ? */; // → u16
    let y = parse_port(b)/* ? */; // → u16
    Ok(x + y)
}
```

```
fn main() {
    // TODO 2: use 8080 when parsing fails (cheap default)
    let p1: u16 = parse_port("9000")/* ? */;

    // TODO 3: build the default lazily (only if needed)
    let p2: u16 = parse_port("oops")/* ? */;

    // TODO 4: fall back to u16::default() (= 0)
    let p3: u16 = parse_port("oops")/* ? */;

    // TODO 5: panic with a message if it must succeed
    let p4: u16 = parse_port("oops")/* ? */;
}
```



Debugging

Find the bug



The compiler is your **first** debugger - then reach for a print.

```
fn main() {  
    let xs = vec![1, 2, 3];  
    println!("xs = {xs:?}"); // you choose what to show  
    let _n = dbg!(xs.len()) * 2; // prints AND returns the value  
}
```

```
Compiling playground v0.0.1 (/playground)  
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.49s  
Running `target/debug/playground`
```

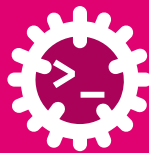
```
[src/main.rs:6:12] xs.len() = 3 ← dbg! prints *file:line*  
                             the *expression* and its *value*
```

```
xs = [1, 2, 3]                ← println! prints what you choose
```

- **println!** - quick, but you **format it yourself**
- **dbg!(expr)** - prints **file:line**, the **expression** and its **value**, then **returns it**
- `rustc --explain E0382` - the **full story** behind an error code



Read the borrow-checker hint first - it usually names the fix.



Beyond a quick print: **levels** you can switch on/off **without recompiling**.

```
use log::{info, warn, error};

fn main() {
    env_logger::init();
    info!("starting on port {}", 8080);
    warn!("connection slow, retrying");
    error!("giving up");
}
```

```
RUST_LOG=info cargo run    # show info and above, everywhere
RUST_LOG=debug cargo run   # show debug and above
RUST_LOG=trace cargo run   # everything
```

- levels: error! > warn! > info! > debug! > trace!
- **log** is a **facade** - pick a backend: **env_logger**, **tracing**
- **tracing** - structured spans, great for **async** / production
- filter via RUST_LOG, **no code change**

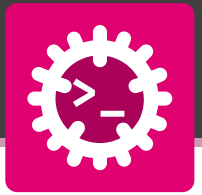


```
===== RUST_LOG=info =====  
[2026-06-23T06:41:25Z INFO logging_levels] application starting  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 1 failed, retrying  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 2 failed, retrying  
[2026-06-23T06:41:25Z INFO logging_levels::net] connected on attempt 3  
[2026-06-23T06:41:25Z WARN logging_levels] disk almost full: 92% used  
[2026-06-23T06:41:25Z ERROR logging_levels] database connection lost  
[2026-06-23T06:41:25Z INFO logging_levels] application stopped
```

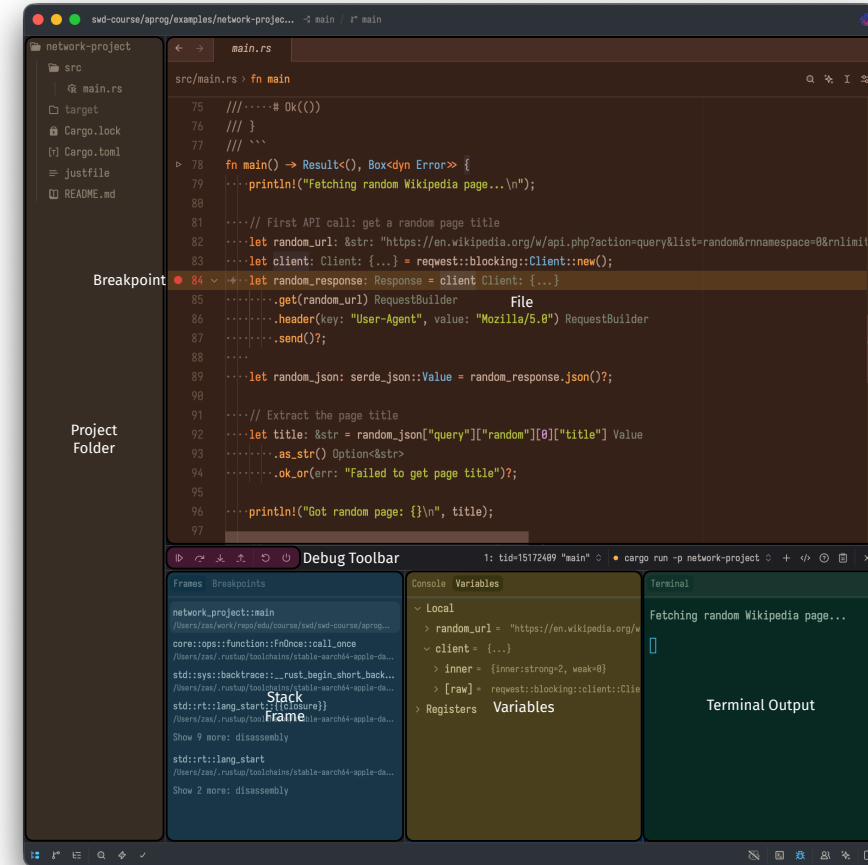
```
===== RUST_LOG=debug =====  
[2026-06-23T06:41:25Z INFO logging_levels] application starting  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 1/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 1 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 2/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 2 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 3/3  
[2026-06-23T06:41:25Z INFO logging_levels::net] connected on attempt 3  
[2026-06-23T06:41:25Z WARN logging_levels] disk almost full: 92% used  
[2026-06-23T06:41:25Z ERROR logging_levels] database connection lost  
[2026-06-23T06:41:25Z INFO logging_levels] application stopped
```

```
===== RUST_LOG=trace =====  
[2026-06-23T06:41:25Z INFO logging_levels] application starting  
[2026-06-23T06:41:25Z TRACE logging_levels::net] entering connect()  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 1/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 1 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 2/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 2 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 3/3  
[2026-06-23T06:41:25Z INFO logging_levels::net] connected on attempt 3  
[2026-06-23T06:41:25Z WARN logging_levels] disk almost full: 92% used  
[2026-06-23T06:41:25Z ERROR logging_levels] database connection lost  
[2026-06-23T06:41:25Z INFO logging_levels] application stopped
```

Debugging with Breakpoints (in Zed / VS Code)



1. Open the *main.rs* file and click the *gutter* to set a *breakpoint*
2. run the program in *debug mode* F4
3. Go through the program with the following actions:
 - *continue* - F5 - run until the next breakpoint
 - *step over* - F7 - run the current line
stop at the next line
 - *step into* - Ctrl+F11 - run the current line
stop at the first line of the called function
 - *step out* - Shift+F11 - run until the current
function returns
4. **Inspect** live the:
 - *stack frame* - see the **call stack**
 - *variables* - see the **current values** of variables in scope
 - *terminal* - see the **program output**



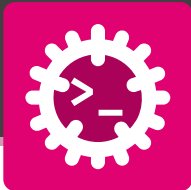


Profiling

Make it fast



Coming up: five **mystery programs** - guess what it does from the time **split** and the **flamegraph**.



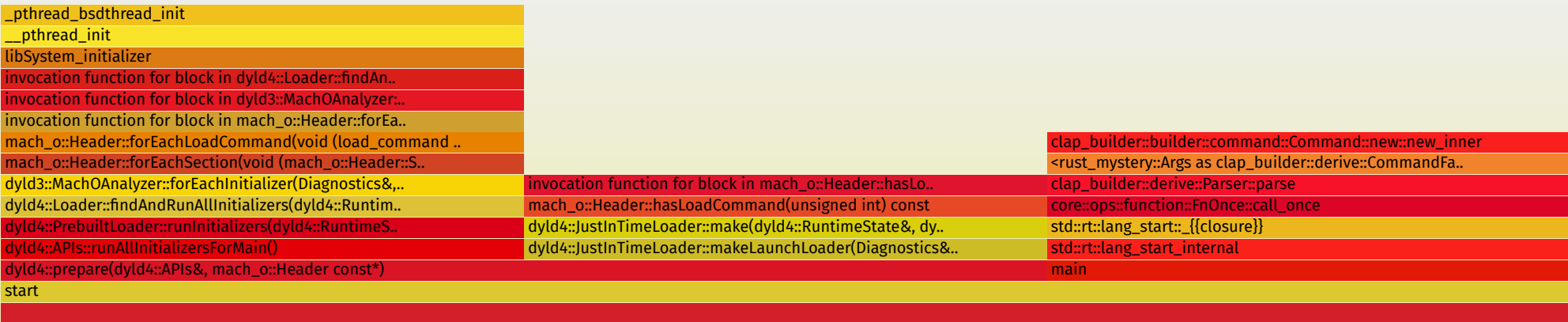
```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 1"
```

Execution Mystery Program 1

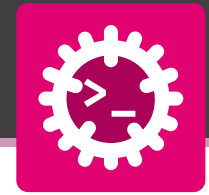
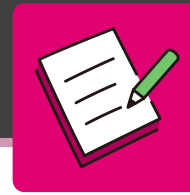
```
...
Time (mean ± σ):      2.013 s ±  0.002 s   [User: 0.003 s, System: 0.004 s]
Range (min ... max):  2.009 s ...  2.014 s   10 runs
```

Flame Graph

Search



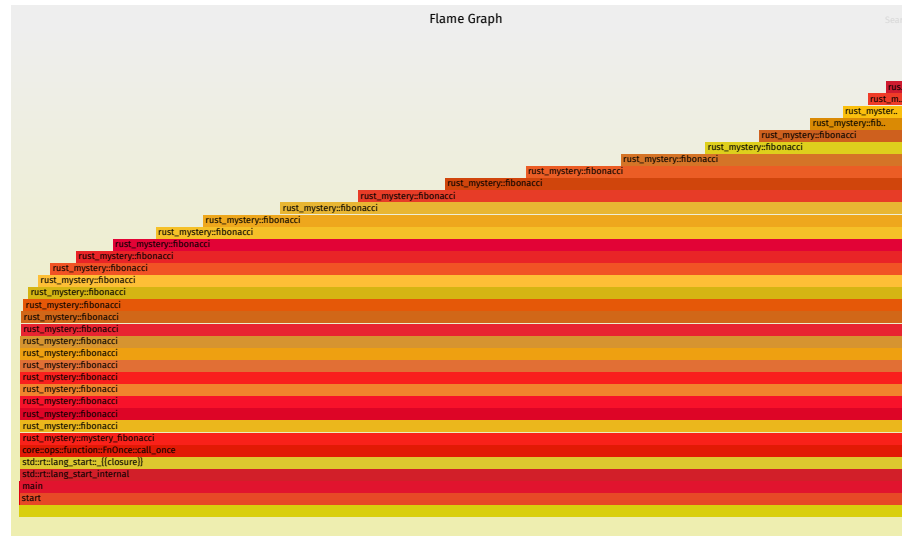
Mystery Program 2



```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 2"
```

Execution Mystery Program 2

```
...  
Time (mean ± σ):      2.213 s ± 0.009 s [User: 2.192 s, System: 0.016 s]  
Range (min ... max):  2.201 s ... 2.227 s   10 runs
```

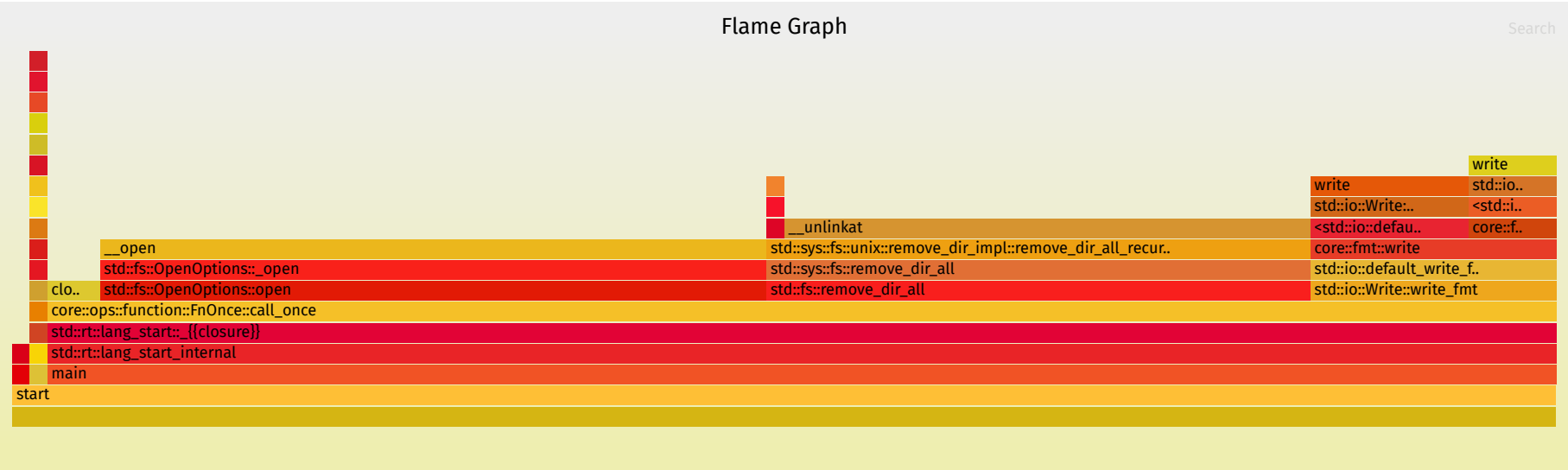




```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 3"
```

Execution Mystery Program 3

```
...
Time (mean ± σ):    115.0 ms ± 18.9 ms   [User: 2.5 ms, System: 107.6 ms]
Range (min ... max): 83.0 ms ... 143.0 ms   24 runs
```





```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 4"
```

Execution Mystery Program 4

```
...
Time (mean ± σ):      760.5 ms ± 45.9 ms   [User: 10451.8 ms, System: 41.7 ms]
Range (min ... max):  703.5 ms ... 852.2 ms   10 runs
```

Flame Graph

Search

```
<core::num::<impl i32>::overflowing_add rust_mystery::mystery_parallel_cpu::_{{closure}}
std::thread::Builder::spawn_unchecked::_{{closure}}::_{{closure}}
std::sys::pal::unix::thread::Thread::new::thread_start
_pthread_start
thread_start
```

Mystery Program 5



```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 5"
```

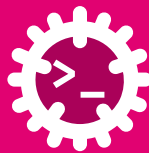
```
Execution Mystery Program 5
results front: 301
results back: 1300
kernel execution duration (ns): 204
...
Time (mean ± σ):      47.4 ms ±   5.6 ms   [User: 11.0 ms, System: 13.5 ms]
Range (min ... max):  41.0 ms ...  79.5 ms   62 runs
```





Documentation

Code others (and future-you) can read

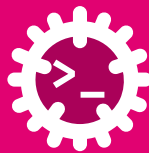


Doc comments become a **browsable website** - and the examples are **compiled and tested**.

```
//! # mycrate
//!
//! Temperature conversions for the AProg course.
//! Crate-level docs (`//!`) sit at the top of `lib.rs`
//! and become the front page of the generated site.
//!
//! # Examples
//! ```
//! use mycrate::temperature::Celsius;
//! assert_eq!(Celsius::new(0.0).to_fahrenheit(), 32.0);
//! ```
/// Temperature handling.
pub mod temperature {
    /// A temperature in degrees Celsius.
    ///
    /// # Examples
    /// ```
    /// use mycrate::temperature::Celsius;
    /// let t = Celsius::new(25.0);
    /// assert_eq!(t.to_fahrenheit(), 77.0);
    /// ```
```

```
pub struct Celsius(f64);
impl Celsius {
    /// Creates a new [`Celsius`] value.
    pub fn new(deg: f64) → Self {
        Self(deg)
    }

    /// Converts to Fahrenheit.
    ///
    /// # Panics
    /// Never - this conversion is total.
    pub fn to_fahrenheit(&self) → f64 {
        self.0 * 9.0 / 5.0 + 32.0
    }
}
```

Documentation lives **next to the code** - the same source is both the comments and the published site.

Writing the comments

- `///` documents the **next item** - a struct, method, or function
- `//!` documents the **enclosing** item - the module or whole crate
- **Markdown** inside: headings, lists, code blocks
- convention sections: `# Examples`, `# Panics`, `# Errors`
- `[Celsius]` - an **intra-doc link** to another item

Building & testing

- the `# Examples` block is **real code**, compiled and run
- `cargo test` executes every doc example → docs **can't drift**
- public crates publish automatically to **docs.rs**

```
# build the HTML docs, open in browser
cargo doc --open

# also compiles & runs the doc examples
cargo test
```

WHY DO WHALES JUMP
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG

WHY ARE THERE MALE AND FEMALE BIKES

WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

WHY ARE THERE
SQUIRRELS



WHY IS THERE HE

WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH

WHY AREN'T THERE DINOSAUR GHOSTS

WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR

WHY DO IGUANAS DIE

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD

WHY ARE THERE ANTS IN MY LAPTOP

WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN

HERE TIME

WHY ARE TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE SWISS AFRAID OF DRAGONS

QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY IS THERE LAVA

WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILEGRIM

WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FUNNY
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE

WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS

WHY ARE THERE FEMALE M

WHY AREN'T
THERE GUNS IN
HARRY POTTER



WHAT IS <https://xkcd.com/1256/>
WHY DO THEY SAY T-MINUS
WHY ARE OCEANS BECOMMING MORE ACIDIC

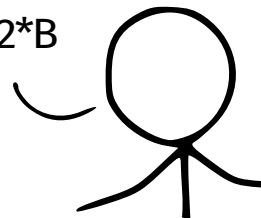
WHY IS LIFE

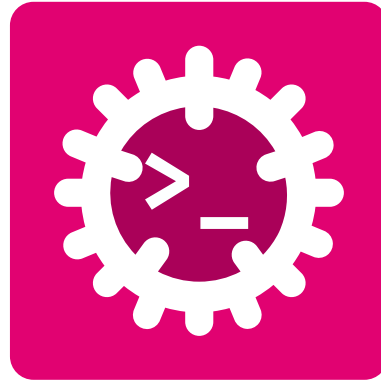
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE

WHY IS YKK ON ALL ZIPPERS

WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTP
WHY IS THERE A RED LINE THROUGH HTTPS ON TWITTE
WHY IS HTTPS IMPORTANT

WHY 2*B | 2*B





Glossary & Bibliography



Computer Science

CPU – Central Processing Unit: The main processing component of a computer. [23](#)

Concurrency

I/O – Input/Output: Operations that involve reading from or writing to external devices or resources. [23](#)



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